

Enrollment No. _____

Ganpat University
Acharya Motibhai Patel Institute of Computer Studies
Dual(BCA-MCA)
First Semester
Subject: U51C4IDE - Introduction to Digital Electronics
First Internal Exam

Date : 14/09/2026 Time: 9:15 AM to 10:15 AM Total: 20 Marks (Theory)

Assessment Pattern

Bloom's Taxonomy Based Assessment Pattern :

BTL1-Remember; BTL2- Understand; BTL3- Apply; BTL4-Analyze; BTL5- Evaluate; BTL6-Create;

Bloom's Category	Section	Description	Marks	Total Marks
BTL1, BTL2	05 question, 1 mark each (Short questions / MCQ/Fill in the blank).	Choose/ Fill ups/ One word, 1-2 Sentence.	05	20
BTL3, BTL4, BTL5, BTL6	03 out of 05 questions, 05 mark each	150-200 Words	15	

Question Paper

Section – A				
Sr.No.	Answer the following	Marks	CO	Bloom's Taxonomy Pattern
1	Define Logic Circuit.	01	CO1	BTL1
2	Which gate is known as the Universal Gate? (a) AND (b) NAND (c) OR (d) EX-OR	01	CO1	BTL1
3	A NOT gate is also called an _____ gate. (a) OR (b) Inverter (c) AND (d) Buffer	01	CO1	BTL2
4	How many inputs and selection lines does a 2-to-1 multiplexer have _____. (a) 2 inputs, 2 selection lines (b) 2 inputs, 1 selection line (c) 1 input, 2 selection lines (d) 4 inputs, 2 selection lines	01	CO2	BTL2
5	Draw logic circuit for the Boolean expression: $Y = A.B + C.D$	01	CO2	BTL3
Section – B				
Answer any 03 from the following				
6	What is Logic Gate? Explain all basic gates.	05	CO1	BTL2
7	State and prove De-Morgan's Theorems.	05	CO1	BTL3
8	Explain Universal Gates with a suitable diagram and truth table.	05	CO1	BTL3
9	Explain Combinational Logic Circuits and Sequential Logic Circuits with examples.	05	CO2	BTL4
10	What is a Multiplexer? Explain its working with truth table and diagram.	05	CO2	BTL3

**** End of the Paper ****